ឯកសារផែនការក្នុងការតេសប្រព័ន្ធសេដ្ឋកិច្ចក្រៅប្រព័ន្ធ

Informal Economy Test Plan Documentation

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I. Overview

Project Name: Informal Economy Software Solution

Objective: To ensure the quality, reliability, and usability of the software

Scope: The testing plan covers aspects of quality assurance, such as manual testing that includes (mobile, client site, agent and admin site), UAT testing, regression testing, functionality testing, API testing, useability testing focusing on meeting the unique needs of the informal economy sector. Stress, Load and Pressure test is also included to analyze the performance of each application with a certain number of users accessing at the same time.

II. Testing Approach

Agile Testing: Align QA activities with agile development methodologies to accommodate changing requirements and deliver incremental value.

Collaborative QA: collaboration between QA engineers, developers, client, and users to ensure an understanding of requirements and expectations.

Continuous Testing: Integrate testing activities throughout the development lifecycle to detect and address issues early, minimizing rework and enhancing product quality.

III. Objectives

- Verify that the software meets functional and non-functional requirements
- Validate usability, accessibility, and user experience (UX) to ensure the software is convenient and user-friendly for participants with diverse backgrounds and skill levels.
- Identify and address defects promptly to deliver a stable and reliable applications

IV. Types of Testing

1. Manual Testing

- **Mobile Testing**

  Objective: Validate the functionality, usability, and performance of the software across different platforms used by participants in the informal economy. Ensure consistency and reliability across mobile, client site, agent site, and admin site.
  Scope: Test the mobile application on different operating systems (iOS, Android), screen sizes, and resolutions.
  Tools: Utilize real devices or emulators for testing. Tools like App tester (Android) or TestFlight (IOS) can be used for automation if applicable.

- **Client Site Testing**

  Objective: Ensure that the client-side components of the software function correctly across various web browsers and versions.
  Scope: Test the user interface, navigation, forms, and interactions on different browsers (Chrome, Firefox, Safari, Edge).
  Tools: Perform manual testing on real browsers or use browser testing tools like BrowserStack or CrossBrowserTesting.

- **Agent Site Testing**

  Objective: Validate the functionality and usability of the agent-side interface used for intermediate interactions.
  Scope: Test agent-specific features, workflows, and data processing capabilities.
  Tools: Perform manual testing on supported browsers and mobiles ensuring compatibility and usability.

- **Admin Site Testing**

  Objective: Ensure that administrative tools and interfaces are robust, secure, and user-friendly for managing the software and its users.
  Scope: Test administrative workflows, permissions, reporting functionalities, and data management features.
  Tools: Conduct manual testing on different browsers/devices, focusing on administrative tasks and responsibilities.

2. UAT Testing (User Acceptance Testing)

Objective: Validate the software's readiness for deployment from the end-users' perspective, ensuring it meets their requirements and expectations.
Scope: Engage real users to execute predefined test cases and scenarios in a production-like environment, focusing on user workflows and use cases.
Tools: Conduct manual testing with the support of PR team

3. Regression Testing

Objective: Ensure that recent code changes do not adversely affect existing functionalities.
Scope: Re-run selected test cases to verify that new code modifications have not introduced defects or regression issues.
Tools: Automate regression tests using tools like Selenium WebDriver

4. API Testing

Objective: Verify the functionality, reliability, and security of APIs used for data exchange and integration.
Scope: Test API endpoints, request/response formats, authentication mechanisms, and error handling.
Tools: Utilize Postman API testing tools

5. Pressure, Stress and Load Testing

Objective: Assess the software's performance, stability, and scalability under various stress conditions, including heavy load and concurrent user activity.
Scope: Simulate high-load scenarios, peak traffic conditions, and resource constraints to identify performance bottlenecks and failure points.
Tools: Utilize load testing tools Apache JMeter

V. Documentation

Test Plan: Document testing strategy, objectives, scope, and schedule.
Test Cases: Create detailed test cases for each testing type, covering various scenarios.
Test Reports: Generate comprehensive test reports to track testing progress and findings.

VI. Schedule

2 months

VII. Continuous Improvement

Feedback Loop: Gather feedback from testers, users, and stakeholders for continuous improvement.
Root Cause Analysis: Analyze defects to identify underlying issues and prevent recurrence.
Knowledge Sharing: Promote knowledge sharing and cross-training

VIII. Conclusion